SMARTSPEND - SMART EXPENSE MANAGER & SPLITWISE CLONE

ABSTRACT

Managing personal finances and shared expenses is an important part of modern life. Traditional methods such as manual records and spreadsheets require additional effort and may cause calculation errors and difficulties in tracking financial activities.

This paper presents SmartSpend, an integrated web-based application that combines personal expense management and group expense splitting. It supports income and expense tracking, category management, budgeting, group creation, shared transactions, automated expense calculations, balance tracking, and settlements.

Developed using Python, Django, and SQLite, SmartSpend provides a centralised platform for organising financial activities, reducing manual effort, and improving financial transparency through dashboard-based analytics.

Keywords: Smart Expense Management, Personal Finance, Expense Splitting, Django, SQLite, Budget Management, Financial Analytics.

I. INTRODUCTION

Personal financial management helps individuals maintain financial stability and develop responsible spending habits. However, traditional methods such as handwritten records and spreadsheets make transaction management, budget monitoring, and spending analysis difficult.

Managing shared expenses among friends, family members, and colleagues creates additional challenges. Different contributions toward common expenses can lead to complicated calculations, confusion, and financial misunderstandings.

Existing applications often provide personal finance management and group expense splitting separately, resulting in fragmented financial records.

To overcome these limitations, SmartSpend proposes a centralised web-based platform that integrates personal expense tracking, budget management, group expense splitting, balance calculation, and settlement management.

The major contributions include:

1. Integration of personal and group financial management.
2. Category-based income and expense tracking.
3. Automated distribution of shared expenses.
4. Individual balance and settlement management.
5. Dashboard-based financial analytics.

II. RELATED WORK

Existing digital financial applications provide various solutions for expense tracking, budgeting, and shared financial management.

A. Personal Expense Management Systems

Traditional financial records require manual calculations and continuous user involvement. Research on

applications such as Expensa and Finexa highlights the importance of digital expense tracking, budget monitoring, financial records, and personalised financial guidance.

B. Group Expense Management Systems

Applications such as Splitwise simplify shared expense management through automated calculations and individual balance tracking. Research on Silent Splitter explores privacy-preserving payment-splitting techniques, while other expense-sharing systems focus on unequal contributions and settlement calculations.

C. Identified Research Gap

Existing systems often handle personal and group financial activities separately. Important opportunities include:

1. Centralised personal and group financial management.
2. Automated expense distribution.
3. Unified financial records.
4. Integrated budget monitoring.
5. Transparent balance and settlement management.

D. Relevance to SmartSpend

SmartSpend combines concepts from personal finance and group expense management research into one platform using Python, Django, and SQLite.

III. PROBLEM STATEMENT AND OBJECTIVES

A. Problem Statement

Managing personal transactions and shared expenses through manual methods creates difficulties in maintaining accurate records, monitoring budgets, calculating individual contributions, and tracking outstanding balances.

Using separate applications for personal and group expenses further increases management effort and fragments financial information.

SmartSpend addresses these challenges through an integrated platform for personal expense tracking, budgeting, group expense splitting, balance calculation, and settlement management.

B. Research Objectives

The primary objective is to develop a web-based application that simplifies personal and shared financial management.

Specific objectives include:

1. Managing personal income and expenses.
2. Creating groups and managing participants.
3. Automating shared expense distribution.
4. Calculating individual contributions and balances.
5. Maintaining settlement records.
6. Monitoring personal budgets.
7. Providing financial summaries and analytics.
8. Integrating all financial activities into one platform.

C. Scope of the Proposed System

SmartSpend covers personal transactions, categories, expense-sharing groups, shared expenses, expense distribution, balance calculations, settlements, budget monitoring, and graphical analytics.

The application uses Python, Django, and SQLite. It focuses on financial record management and calculations. Banking integration and online payment processing are outside its current scope, and transactions must be entered manually.

IV. PROPOSED SYSTEM AND METHODOLOGY

A. Overview of the Proposed System

SmartSpend is a modular web application that integrates personal expense tracking and group expense management through a centralised Django backend and SQLite database.

B. Proposed System Modules

The system consists of nine major modules:

1. User Authentication: Registration, login, logout, sessions, and access control.
2. Personal Expense Management: CRUD operations for income, expenses, and categories.
3. Group Management: Group creation and participant management.
4. Group Expense Management: Recording shared transactions and contributors.
5. Expense Splitting: Automatic distribution of expenses among participants.
6. Balance and Settlement Management: Calculating individual balances and recording settlements.
7. Budget Management: Defining spending limits and monitoring expenditure.
8. Dashboard and Analytics: Displaying financial summaries and graphical reports.
9. Database Management: Managing application records using SQLite.

The equal expense distribution formula is:

$$S = \frac{E}{N}$$

Where:

- S = Individual expense share.
- E = Total group expense.
- N = Number of participating members.

For example, ₹2,000 divided among four members results in ₹500 per member.

The net balance formula is:

$$B_i = P_i - S_i$$

Where:

- B_i = Net balance.
- P_i = Total amount paid.
- S_i = Allocated expense share.

A positive balance indicates excess contribution, while a negative balance indicates an outstanding amount.

C. System Development Methodology

The development process follows an iterative methodology involving:

1. Requirement Analysis – Identifying functional and technical requirements.
2. System Design – Designing architecture, modules, and interfaces.
3. Database Design – Defining entities and relationships.

4. Application Development – Implementing backend and frontend functionality.
5. Integration – Connecting modules with Django and SQLite.
6. Testing and Validation – Verifying application functionality and calculations.
7. Deployment Preparation – Reviewing configuration and execution requirements.

D. Expense Splitting Methodology

The expense-splitting process involves recording the total expense, identifying participants, selecting a distribution method, calculating individual shares, recording contributions, determining balances, and updating settlement records.

E. Data Management and Processing

SmartSpend uses SQLite for structured storage and Django ORM for database operations. Personal and group transactions are maintained through interconnected records. Authentication, input validation, and access control help maintain data consistency and security.

F. Expected Outcomes

The system aims to simplify expense tracking, automate shared calculations, improve settlement transparency, support budget monitoring, and provide centralised financial analytics.

V. SYSTEM ARCHITECTURE AND DESIGN

A. System Architecture

SmartSpend follows a three-layer architecture:

1. Presentation Layer: HTML5, CSS3, JavaScript, and Bootstrap 5 provide responsive interfaces for financial activities.
2. Application Layer: Python and Django handle authentication, URL routing, transaction processing, expense calculations, budgeting, and settlements.
3. Database Layer: SQLite stores users, transactions, categories, groups, participants, expenses, budgets, and settlements through Django ORM.

B. System Architecture Diagram

Figure 1. Overall System Architecture of SmartSpend

The frontend communicates with the Django backend, which processes requests and interacts with SQLite to retrieve or store financial information.

C. Module Architecture

The nine interconnected modules operate through the central Django backend and share relevant information through the database.

Figure 2. Module Architecture of SmartSpend

D. Database Design

The database contains nine major entities:

1. User
2. Category
3. Personal Transaction
4. Group
5. Group Member
6. Group Expense
7. Expense Split
8. Budget
9. Settlement

E. Database Relationship Design

The database uses primary keys and foreign keys to maintain relationships. A user can manage multiple transactions, budgets, and groups. Groups contain multiple members and expenses, while shared expenses generate individual split records. Settlements are associated with relevant participants and group activities.

Figure 3. Entity Relationship Diagram of SmartSpend

F. User Interface Design

The dashboard-oriented interface provides access to:

- Dashboard and financial summaries.
- Personal income and expenses.
- Groups and participants.
- Shared group expenses.
- Budget monitoring.
- Settlements and balances.
- Graphical financial analytics.

Responsive interfaces are developed using HTML5, CSS3, JavaScript, and Bootstrap 5.

G. System Design Considerations

The system focuses on modularity, scalability, maintainability, data consistency, security, and responsive design.

VI. SYSTEM IMPLEMENTATION

A. Development Environment

The application uses the following technologies:

1. Python – Backend logic and financial calculations.
2. Django – Application framework and request processing.
3. SQLite – Database management.
4. HTML5 and CSS3 – Frontend structure and styling.
5. JavaScript – Interactive functionality.
6. Bootstrap 5 – Responsive interface components.
7. Chart.js – Financial data visualisation.

B. Implementation of Core Modules

The application implements authentication, personal transaction CRUD operations, group creation, participant management, shared expense recording, automated expense splitting, balance calculation, settlements, budget monitoring, and dashboard analytics through the Django backend.

C. Database Implementation Using Django ORM

Django ORM manages database operations through Python models. It supports creating, retrieving, updating, and deleting records while maintaining relationships between application entities. Database migrations maintain the database structure.

D. Security Implementation

The system incorporates:

1. User authentication and session management.
2. Password hashing through Django.
3. Form validation.
4. CSRF protection.
5. User-specific data access restrictions.

6. Database integrity through model relationships and constraints.

E. Implementation Outcome

The implementation integrates personal expense tracking, group expense sharing, automated balance calculations, budget monitoring, and financial visualisation within a centralised web application.

VII. RESULTS AND DISCUSSION

A. System Implementation Results

SmartSpend was developed using Python, Django, and SQLite. It integrates personal financial management and group expense-sharing functionalities into one platform.

B. Functional Results

The implemented system supports:

1. User registration, login, and protected access.
2. Personal income and expense management.
3. Group creation and participant management.
4. Shared transaction recording.
5. Automated expense distribution.
6. Balance calculation and settlement tracking.
7. Budget monitoring.
8. Dashboard-based financial analytics.

C. Expense Splitting Result Analysis

Consider four members sharing an expense of ₹2,000 equally.

$$S = \frac{2000}{4} = ₹500$$

Each member is responsible for ₹500. If one member pays the entire amount, that member contributes ₹1,500 more than their allocated share, while the remaining three members owe ₹500 each.

This demonstrates how automated expense calculations simplify shared financial management.

D. Testing and Validation

Testing focuses on authentication, personal transaction CRUD operations, group management, expense distribution, balance calculations, settlement updates, budget monitoring, and dashboard functionality.

E. Discussion

The implementation demonstrates how web technologies can integrate personal and group financial management. Automated calculations reduce manual effort, while budget monitoring and graphical summaries improve financial organisation.

However, system accuracy depends on correctly entered information because transactions and settlements are recorded manually.

F. Limitations of the System

1. Manual financial transaction entry.
2. No automatic bank synchronisation.
3. No online payment processing.
4. Dependence on accurate user-entered information.
5. SQLite may require replacement for large-scale deployment.

VIII. CONCLUSION AND FUTURE SCOPE

A. Conclusion

This research presented SmartSpend, an integrated web-based system for personal expense management and group expense splitting.

Developed using Python, Django, and SQLite, the application combines authentication, personal transactions, group management, expense splitting, balance calculations, settlements, budgeting, and dashboard analytics.

The system provides a centralised environment for organising financial records, monitoring expenditures, and managing shared expenses. Its modular architecture also supports future improvements and enhancements.

B. Future Scope

Future enhancements may include:

1. AI-Based Financial Insights: Analysing spending patterns and generating personalised insights.
2. Smart Budget Recommendations: Suggesting category-wise budgets using historical expenditure.
3. Bank Integration: Synchronising transactions through financial APIs.
4. Online Payments: Integrating payment gateways for group settlements.
5. Mobile Applications: Developing Android and iOS applications.
6. Advanced Analytics: Providing detailed financial trends and comparisons.
7. Cloud Deployment: Improving scalability and accessibility.
8. Enhanced Security: Introducing multi-factor authentication and stronger access controls.

C. Final Remarks

SmartSpend establishes a foundation for centralised personal and group financial management. Future enhancements can improve its intelligence, accessibility, security, and scalability.

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